



Vojtěch Špaček

Senior C++ engineer — safety-critical embedded & HIL test automation | Linux/networking | agentic-AI (MCP) tooling

Pragmatic programmer and creative problem solver — systems, verification and open source.

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Nationality: EU (Czech Republic) **Place of residence:** Brno, CZ

Work permit: EU citizen (Czech Republic) — no permit required in the EU/EEA. B-permit eligible.

Profile Summary

Senior C++ engineer working on safety-critical and real-time systems: aircraft onboard diagnostics at Honeywell, electron-microscopy detector software at Thermo Fisher Scientific, and a Linux inspection proxy and firewall at Trusted Network Solutions.

Most recently sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), raising measured conformance to 259 of 271 tracked state transitions (95.6%).

Linux, networking and security specialist — Netlink, nftables, TLS inspection, Kerberos/LDAP/OCSP — with C++20 daemons shipped as Debian packages and a Rust reimplementation published on crates.io.

Builds agentic-AI tooling on the Model Context Protocol: an agent harness adopted into a daily client delivery workflow, and MCP-Prompts, published to npm, Docker Hub and the MCP directories (Glama, MCP Market, PulseMCP) with 97 stars and 21 forks.

Work Experience

Test Automation Engineer (Python, C#)

Resideo (Ademco) (via CoolPeople (contractor))

Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), as sole owner of the verification test suite and of the automated bench that drives it. Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

- Designed an Arrange/Act/Assert validation pipeline in Python with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator expanding a YAML catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.
- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering a pyvisa/USBTMC/SCPI programmable AC source, CAN serial-bus monitoring and Modbus relay and probe I/O.

Software Developer (C++)

Thermo Fisher Scientific (via Ixperta (contractor)) · 2023 – 2024

Electron microscopy software for high-resolution detector hardware operation and image acquisition. Built server-side C++ applications, FPGA-based data processing pipelines with gRPC networking, and backend logic for a C#/NET WPF image processing and display application. Worked in an international Agile team with bi-weekly stakeholder demos.

- Implemented server-side software in C++, including hardware abstraction layers (HAL) and inter-process communication (IPC) using COM and MIDL on Windows.
- Developed embedded software for FPGA hardware, using gRPC network communication in C++, Python and Linux environments.
- Optimized the data acquisition pipeline and implemented robust error handling.

Software Engineer II

Honeywell · 2022 – 2023

Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight. Architected and implemented the DiagTest framework, a state machine-based diagnostic test engine in C++ managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

- Designed a job-based execution system with DiagTestJob and DiagTestMsgQueueProc for asynchronous test orchestration, implementing message queue processing patterns for reliable real-time diagnostics.
- Engineered a test interference management system preventing conflicting diagnostic tests from running simultaneously, ensuring aircraft system safety and data integrity.
- Implemented four diagnostic test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.

Application Developer

Trusted Network Solutions · 2020 – 2022

Kernun Adaptive Firewall. Implemented a forward proxy in C++ on Linux featuring HTTP/HTTPS proxying with TLS/SSL inspection, OCSP integration for certificate validation, Kerberos authentication and LDAP user/group lookup, plus the ipmon network monitoring daemon driving nftables firewall rules from real-time IPv4/IPv6 address changes.

- Built comprehensive proxy infrastructure with certificate generation and management (CA, client certificates), a dynamic configuration builder/parser system, and a manager/partition architecture for modular daemon design.
- Integrated security components including Suricata (intrusion detection) and nftables (static firewall), aggregating logs and managing connection information in PostgreSQL.
- Implemented real-time network interface monitoring in ipmon using NETLINK sockets, handling both IPv4 and IPv6 address management, with Unix domain sockets for IPC and dynamic nftables rule management.

Independent IT Contractor

Self-employed

Independent contracting for business and consumer clients: CI/CD automation pipelines and Linux server administration for B2B customers, and IT support, consumer hardware retail and security consulting for B2C customers.

- B2B: CI/CD automation pipelines and Linux server administration.
- B2C: IT support freelancing, consumer hardware retail and security consulting.

Automation Developer & IT Systems Specialist

IBM · 2014 – 2016

Enterprise IT operations: infrastructure monitoring, incident management and an automation solution that replaced manual ticketing and server administration work.

- Developed an automation application that replaced manual workload in ticketing operations and basic server administration, using Bash, PowerShell and JavaScript for cross-platform server management and applying automata theory to design state-driven workflows.
- Monitored Unix/Windows servers and resolved production incidents, working hands-on with Active Directory and VMware virtualization.

Skills

Embedded & RTOS: C++ (C++17/C++20) (Senior) · Embedded software / ARM / RTOS (Medior) · State machine design patterns (Senior) · Hardware abstraction layers & IPC (COM/MIDL, Unix sockets) (Medior) · FPGA-adjacent development & gRPC (Medior) · Flatbuffers (binary serialization) (Medior) · Qt/QML (Working knowledge)

Verification & HIL: Python (Medior) · Hardware-in-the-loop test design (Senior) · pyvisa / USBTMC / SCPI instrument control (Medior) · Modbus I/O cards & CAN serial-bus diagnostics (Medior) · Safety-certified product verification (UL/CSA, ANSI Z21.20, NFPA 54) (Medior) · Requirement traceability & test evidence reporting (openpyxl, HTML, Mermaid) (Medior) · C# / .NET (Medior) · Perl test frameworks (Medior)

Networking & Security: Linux (Senior) · Linux networking, TCP/IP, Netlink, nftables (Senior) · Encrypted traffic inspection (TLS 1.3, OCSP) (Medior) · Kerberos/Heimdal, LDAP, PostgreSQL, Suricata IDS (Medior) · Forward/TCP/HTTP proxy design (Senior) · Rust (Working knowledge)

AI / LLM / MCP: Model Context Protocol server development (Senior) · Agentic workflow design & governance (agents, skills, policy hooks) (Medior) · Prompt engineering (RAG, few-shot, prompt caching, structured outputs, tool calling) (Medior) · LLM provider APIs & SDK integration (Medior) · JavaScript / TypeScript (Medior) · Node.js (Medior)

Tooling: Git (Senior) · Bash (Senior) · Docker / Podman (Medior) · Conan (C/C++ dependency management) (Medior) · CI/CD (Jenkins, TeamCity, GitLab CI, GitHub Actions) (Medior) · PowerShell (Medior) · Enterprise Architect (UML) (Working knowledge) · GDB, Wireshark, VirtualBox (Medior)

Open Source & Projects

MCP-Prompts — 97 stars, 21 forks | github.com/sparesparrow/mcp-prompts

TypeScript · Node.js · PostgreSQL + pgvector · Docker · REST + SSE · AWS

rust-network-mgr | github.com/sparesparrow/rust-network-mgr

Rust · Netlink · nftables · epoll · Lock-free data structures

rtp-midi — 2 stars | github.com/sparesparrow/rtp-midi

Rust (66.5%) · RTP-MIDI · OSC · Android NDK (Kotlin) · ESP32 / FreeRTOS · Docker

Education & Training

Masaryk University, Faculty of Informatics — Computer Science

Coursework only — no academic title — Coursework in programming, Linux and networking.

Coursera.org / DeepLearning.AI — Machine learning and LLM engineering

Coursework only — no academic title — Deep Learning Specialization (four machine learning courses by Andrew Ng), plus data analysis, prompt engineering and building LLM agents.

Self-directed study — Design patterns, software architecture and offensive security

Coursework only — no academic title — Design Patterns (refactoring.guru) and Hack The Box cybersecurity training; self-studied the Certified Ethical Hacker syllabus (exam not taken).

Professional development reading — Software engineering practice

Coursework only — no academic title — Software Engineering at Google, The Pragmatic Programmer, Clean Architecture.

Languages

Czech (native) · Slovak (native) · English (C1) · German (A1)

References

References available on request — two to three referees and work certificates (Arbeitszeugnisse) can be provided on request.

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